

Title: Problem Sheet 1
Course: AERO40006 (Ordinary Differential Equations)
Lecturer: Peter Vincent

Question 1: Identify the order of the following ODEs, also identify whether they are linear or non-linear.

(a) $\cos(2x)\frac{d^4y}{dx^4} + x = y$

(c) $(x^2 - 1)\frac{d^2y}{dx^2} - \sin(x)\frac{dy}{dx} + xy = x^2$

(b) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right) = 1$

(d) $\frac{dy}{dx} + y = x + y^2$

(a) 4th order
linear

(b) 2nd order
non-linear

(c) 2nd order
linear

(d) 1st order
non-linear

Question 2: Find particular solutions of the following simple first-order ODEs

(a) $\frac{dy}{dx} = x$ $y(0) = 2$

(c) $\frac{dy}{dx} = \cos(x)$ $y(0) = 1$

(b) $\frac{dy}{dx} = e^x$ $y(0) = 3$

(d) $\frac{dy}{dx} = \tan(x)$ $y(0) = 4$

(a) $\int dy = \int x dx$

$$y = \frac{1}{2}x^2 + C$$

When $y(0) = 2$,

$$2 = 0 + C$$

$$\therefore C = 2$$

$$\therefore y = \frac{1}{2}x^2 + 2$$

(b) $\int dy = \int e^x dx$

$$y = e^x + C$$

When $y(0) = 3$

$$1 = 1 + C$$

$$\therefore C = 2$$

$$\therefore y = e^x + 2$$

Question 2: Find particular solutions of the following simple first-order ODEs

(a) $\frac{dy}{dx} = x$ $y(0) = 2$

(c) $\frac{dy}{dx} = \cos(x)$ $y(0) = 1$

(b) $\frac{dy}{dx} = e^x$ $y(0) = 3$

(d) $\frac{dy}{dx} = \tan(x)$ $y(0) = 4$

(c) $\int dy = \int \cos(x) dx$

$$y = \sin(x) + C$$

When $y(0) = 1$,

$$1 = 0 + C$$

$$\therefore C = 1$$

$$\therefore y = \sin(x) + 1$$

(d) $\int dy = \int \tan(x) dx$

$$= \int \frac{\sin(x)}{\cos(x)} dx$$

Let $u = \cos(x)$,

$$\frac{du}{dx} = -\sin(x)$$

$$\therefore dx = \frac{du}{-\sin(x)}$$

$$\therefore y = \int -\frac{1}{u} du$$

$$= -\ln|u| + C$$

$$= -\ln|\cos(x)| + C$$

When $y(0) = 4$,

$$4 = 0 + C \quad \therefore C = 4$$

$$\therefore y = -\ln|\cos(x)| + 4$$

Question 3: Find particular solutions of the following separable first-order ODEs

(a) $\frac{dy}{dx} = (1+x)(1+y)$ $y(0) = 0$

(c) $x \frac{dy}{dx} = y + xy$ $y(1) = 1$

(b) $\frac{dy}{dx} = \frac{1+y}{2+x}$ $y(0) = 1$

(d) $\frac{dy}{dx} = 8x^3y^2$ $y(2) = 3$

(a) $\int \frac{1}{1+y} dy = \int (1+x) dx$

(b) $\int \frac{1}{1+y} dy = \int \frac{1}{2+x} dx$

$$\ln|1+y| = x + \frac{1}{2}x^2 + A$$

$$\ln|1+y| = \ln|2+x| + A$$

When $y(0) = 0$,

When $y(0) = 1$,

$$A = 0$$

$$A = 0$$

$$\therefore \ln|1+y| = x + \frac{1}{2}x^2$$

$$\therefore \ln|1+y| = \ln|2+x|$$

$$\begin{aligned}\therefore 1+y &= e^{x + \frac{1}{2}x^2} \\ &= e^x e^{\frac{1}{2}x^2}\end{aligned}$$

$$\therefore e^{\ln|1+y|} = e^{\ln|2+x|}$$

$$\therefore |1+y| = |2+x|$$

$$\therefore y = e^x e^{\frac{1}{2}x^2} - 1$$

$$\therefore |y| = |x+1|$$

$$\therefore y = x+1$$

Question 3: Find particular solutions of the following separable first-order ODEs

(a) $\frac{dy}{dx} = (1+x)(1+y)$ $y(0) = 0$

(c) $x \frac{dy}{dx} = y + xy$ $y(1) = 1$

(b) $\frac{dy}{dx} = \frac{1+y}{2+x}$ $y(0) = 1$

(d) $\frac{dy}{dx} = 8x^3y^2$ $y(2) = 3$

(c) $\int \frac{1+x}{x} dx = \int \frac{1}{y} dy$

(d) $\int \frac{1}{y^2} dy = \int 8x^3 dx$

$\ln|y| = \ln|x| + x + C$

$-\frac{1}{y} = 2x^4 + C$

When $y(1) = 1$,

When $y(2) = 3$

$C = -1$

$-\frac{1}{3} = 32 + C$

$\therefore \ln|y| = \ln|x| + x - 1$

$\therefore C = -\frac{97}{3}$

$\therefore e^{\ln|y|} = e^{\ln|x| + x - 1}$

$\therefore y = \frac{-1}{2x^4 + \frac{97}{3}}$

$\therefore |y| = e^{\ln|x|} e^x e^{-1}$
 $= |x| e^x e^{-1}$

$\therefore y = \frac{-3}{6x^4 + 97}$

$\therefore y = x e^{(x-1)}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

$$(a) \frac{dy}{dx} = \frac{x^2 + y^2}{xy} \quad y(1) = 0$$

$$(c) x \frac{dy}{dx} = \frac{y^2}{x} + y \quad y(1) = 1$$

$$(b) 2x^2 \frac{dy}{dx} = x^2 + y^2 \quad y(1) = 0$$

$$(d) x^3 \frac{dy}{dx} = x^2 y - 2y^3 \quad y(1) = 6$$

$$(a) \text{ Let } u = \frac{y}{x} \quad \therefore y = ux \quad \therefore \frac{dy}{dx} = \frac{du}{dx}x + u$$

$$\therefore \frac{du}{dx} = \frac{x}{y} + \frac{y}{x} = \frac{1}{u} + u = x \frac{du}{dx} + u$$

$$\therefore x \frac{du}{dx} = \frac{1}{u}$$

$$\therefore \int \frac{1}{x} dx = \int u du$$

$$\therefore \ln|x| = \frac{1}{2}u^2 + C$$

$$\therefore \ln|x| = \frac{1}{2}\left(\frac{y}{x}\right)^2 + C$$

$$\text{When } y(1) = 0, C = 0$$

$$\therefore \ln|x| = \frac{1}{2}\left(\frac{y}{x}\right)^2 = \frac{y^2}{2x^2}$$

$$\therefore y^2 = 2x^2 \ln|x|$$

$$\therefore y = x \sqrt{2 \ln|x|} = x \sqrt{\ln(x^2)}$$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

$$(a) \frac{dy}{dx} = \frac{x^2 + y^2}{xy} \quad y(1) = 0$$

$$(c) x \frac{dy}{dx} = \frac{y^2}{x} + y \quad y(1) = 1$$

$$(b) 2x^2 \frac{dy}{dx} = x^2 + y^2 \quad y(1) = 0$$

$$(d) x^3 \frac{dy}{dx} = x^2 y - 2y^3 \quad y(1) = 6$$

$$(b) \frac{dy}{dx} = \frac{1}{2} + \frac{1}{2} \left(\frac{y}{x} \right)^2$$

$$\text{Let } u = \frac{y}{x} \quad \therefore y = ux \quad \therefore \frac{dy}{dx} = x \frac{du}{dx} + u$$

$$\therefore x \frac{du}{dx} + u = \frac{1}{2} + \frac{1}{2} u^2$$

$$\therefore \int \frac{2}{u^2 - 2u + 1} du = \int \frac{1}{x} dx$$

$$\therefore \int \frac{2}{(u-1)^2} du = \ln|x| + A$$

$$\therefore \frac{-2}{(u-1)} = \ln|x| + B$$

$$\therefore \frac{-2}{\left(\frac{y}{x}-1\right)} = \ln|x| + B$$

$$\text{When } y(1) = 0, B = -2$$

$$\therefore -2 = \frac{y}{x} \ln|x| - \ln|x| + \frac{2y}{x} - 2 \quad \therefore x \ln|x| = y(\ln|x| + 2)$$

$$\therefore -2 = \ln|x| - \frac{2y}{x} - 2 \quad \therefore -2x = x \ln|x| - 2y - 2$$

$$\therefore y = \frac{1}{2} x \ln|x| + x - 1 \quad \therefore y = \frac{x \ln|x|}{\ln|x| + 2}$$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

(a) $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ $y(1) = 0$

(c) $x \frac{dy}{dx} = \frac{y^2}{x} + y$ $y(1) = 1$

(b) $2x^2 \frac{dy}{dx} = x^2 + y^2$ $y(1) = 0$

(d) $x^3 \frac{dy}{dx} = x^2y - 2y^3$ $y(1) = 6$

(c) $\frac{dy}{dx} = \frac{y^2}{x^2} + \frac{y}{x}$

Let $u = \frac{y}{x}$ $\therefore y = ux$ $\therefore \frac{dy}{dx} = \frac{du}{dx}x + u$

$\therefore x \frac{du}{dx} + u = u^2 + u$

$\therefore x \frac{du}{dx} = u^2$

$\therefore \int \frac{1}{u^2} du = \int \frac{1}{x} dx$

$\therefore -\frac{1}{u} = \ln|x| + C$

$\therefore -\frac{x}{y} = \ln|x| + C$

When $y(1) = 1$, $C = -1$ $\therefore x = (1 - \ln|x|)y$

$\therefore y = \frac{x}{1 - \ln|x|}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

$$(a) \frac{dy}{dx} = \frac{x^2 + y^2}{xy} \quad y(1) = 0$$

$$(c) x \frac{dy}{dx} = \frac{y^2}{x} + y \quad y(1) = 1$$

$$(b) 2x^2 \frac{dy}{dx} = x^2 + y^2 \quad y(1) = 0$$

$$(d) x^3 \frac{dy}{dx} = x^2 y - 2y^3 \quad y(1) = 6$$

$$(d) \frac{dy}{dx} = \frac{y}{x} - 2\left(\frac{y}{x}\right)^3$$

$$\text{Let } u = \frac{y}{x} \quad \therefore y = ux \quad \therefore \frac{dy}{dx} = \frac{du}{dx}x + u$$

$$\therefore \frac{du}{dx}x + u = u - 2u^3$$

$$\therefore \frac{du}{dx}x = -2u^3$$

$$\therefore \int -\frac{1}{2u^3} du = \int \frac{1}{x} dx$$

$$\therefore \frac{1}{4u^2} = \ln|x| + C$$

$$\therefore \frac{x^2}{4y^2} = \ln|x| + C$$

$$\text{When } y(1) = 6, \quad C = \frac{1}{144} \quad \therefore \frac{x^2}{4} = (\ln|x| + \frac{1}{144})y^2$$

$$\therefore x^2 = (4\ln|x| + \frac{1}{36})y^2 \quad \therefore y = \frac{x}{2\sqrt{\ln|x| + \frac{1}{144}}}$$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\frac{dy}{dx} + 5y = e^{2x}$ $y(0) = 0$

(c) $\frac{dy}{dx} + \frac{y}{x} = 3x^2$ $y(1) = 5$ $x > 0$

(b) $x \frac{dy}{dx} - 4y - x^7 = 0$ $y(1) = 0$

(a) Let $I = e^{\int 5 dx} = e^{5x}$ $\therefore \frac{dI}{dx} = 5e^{5x}$

$$e^{5x} \frac{dy}{dx} + 5e^{5x} y = e^{5x} e^{2x} = e^{7x}$$

$$\therefore I \frac{dy}{dx} + \frac{dI}{dx} y = e^{7x}$$

$$\therefore \frac{dIy}{dx} = e^{7x}$$

$$\therefore Iy = \int e^{7x} dx = \frac{1}{7} e^{7x} + C \quad \therefore e^{5x} y = \frac{1}{7} e^{7x} + C$$

When $y(0) = 0$

$$\therefore C = -\frac{1}{7}$$

$$\therefore e^{5x} y = \frac{1}{7} e^{7x} - \frac{1}{7}$$

$$\therefore y = \frac{1}{7} e^{2x} - \frac{1}{7} e^{-5x}$$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\frac{dy}{dx} + 5y = e^{2x}$ $y(0) = 0$

(c) $\frac{dy}{dx} + \frac{y}{x} = 3x^2$ $y(1) = 5$ $x > 0$

(b) $x \frac{dy}{dx} - 4y - x^7 = 0$ $y(1) = 0$

(b) $\frac{dy}{dx} - 4\frac{y}{x} - x^6 = 0$

$$I = e^{\int -\frac{4}{x} dx} = e^{-4 \ln|x|} = e^{\ln|x|^{-4}} = x^{-4}$$

$$\therefore \frac{dI}{dx} = -4x^{-5}$$

$$\therefore x^{-4} \frac{dy}{dx} - 4x^{-5}y = x^2$$

$$\therefore I \frac{dy}{dx} - \frac{dI}{dx} y = x^2$$

$$\therefore \frac{dIy}{dx} = x^2$$

$$\therefore Iy = \int x^2 dx \quad \therefore Iy = \frac{1}{3}x^3 + C \quad \therefore x^{-4}y = \frac{1}{3}x^3 + C$$

$$\text{When } y(1) = 0, \quad C = -\frac{1}{3}$$

$$\therefore x^{-4}y = \frac{1}{3}x^3 - \frac{1}{3}$$

$$\therefore y = \frac{1}{3}x^7 - \frac{1}{3}x^4$$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\frac{dy}{dx} + 5y = e^{2x}$ $y(0) = 0$

(c) $\frac{dy}{dx} + \frac{y}{x} = 3x^2$ $y(1) = 5$ $x > 0$

(b) $x \frac{dy}{dx} - 4y - x^7 = 0$ $y(1) = 0$

(c) $I = e^{\int \frac{1}{x} dx} = e^{\ln|x|} = x$

$\therefore \frac{dI}{dx} = 1$

$\therefore x \frac{dy}{dx} + y = 3x^3$

$\therefore I \frac{dy}{dx} + \frac{dI}{dx} y = 3x^3$

$\therefore \frac{dIy}{dx} = 3x^3$

$\therefore Iy = \int 3x^3 dx = \frac{3}{4}x^4 + C$

$\therefore y = \frac{3}{4}x^3 + \frac{C}{x}$

When $y(1) = 5$, $5 = \frac{3}{4} + C \therefore C = \frac{17}{4}$

$\therefore y = \frac{3}{4}x^3 + \frac{17}{4x}$

Question 6: Find general implicit solutions of the following exact first-order ODEs

$$(a) \frac{dy}{dx} = \frac{-\cos(xy) + xy \sin(xy)}{-x^2 \sin(xy) + 2y}$$

$$(b) \frac{dy}{dx} = \frac{3x^2 - 2xy}{x^2 - 2y}$$

$$(a) f(x, y) = -\cos(xy) + xy \sin(xy)$$

$$g(x, y) = x^2 \sin(xy) - 2y$$

$$\therefore \frac{df}{dy} = x \sin(xy) + x \sin(xy) + x^2 y \cos(xy) = 2x \sin(xy) + x^2 y \cos(xy)$$

$$\frac{dg}{dx} = 2x \sin(xy) + x^2 y \cos(xy) \quad \therefore \frac{df}{dy} = \frac{dg}{dx}$$

$$\text{Let } \frac{du}{dx} = f, \quad \frac{dv}{dy} = g$$

$$\begin{aligned} \therefore u_1 &= \int f dx = \frac{1}{y} \sin(xy) - x \cos(xy) - \frac{1}{y} \sin(xy) + p(y) \\ &= -x \cos(xy) + p(y) \end{aligned}$$

$$u_2 = \int g dy = -x \cos(xy) - y^2 + q(y)$$

$$\therefore \frac{du_1}{dy} = x^2 \sin(xy) + \frac{dp(y)}{dy} = x^2 \sin(xy) - 2y$$

$$\therefore p(y) = \int -2y dy = -y^2 + A$$

Question 6: Find general implicit solutions of the following exact first-order ODEs

$$(a) \frac{dy}{dx} = \frac{-\cos(xy) + xy \sin(xy)}{-x^2 \sin(xy) + 2y}$$

$$(b) \frac{dy}{dx} = \frac{3x^2 - 2xy}{x^2 - 2y}$$

$$\therefore \frac{du_2}{dx} = -\cos(xy) + xy \sin(xy) + \frac{dq_2}{dx} = xy \sin(xy) - \cos(xy)$$

$$\therefore q'(y) = 0$$

$$\therefore u = -x \cos(xy) - y^2 + C$$

Question 6: Find general implicit solutions of the following exact first-order ODEs

$$(a) \frac{dy}{dx} = \frac{-\cos(xy) + xy \sin(xy)}{-x^2 \sin(xy) + 2y}$$

$$(b) \frac{dy}{dx} = \frac{3x^2 - 2xy}{x^2 - 2y}$$

$$(b) f = 3x^2 - 2xy$$

$$g = -x^2 + 2y$$

$$\therefore \frac{df}{dy} = -2x, \quad \frac{dg}{dx} = -2x \quad \therefore \frac{df}{dy} = \frac{dg}{dx}$$

$$\therefore f = \frac{du_1}{dx}, \quad g = \frac{du_2}{dy}$$

$$\therefore u_1 = \int 3x^2 - 2xy \, dx = x^3 - x^2y + p$$

$$u_2 = \int -x^2 + 2y \, dy = -x^2y + y^2 + q$$

$$\therefore \frac{du_1}{dy} = -x^2 + \frac{dp}{dy} = -x^2 + 2y \quad \therefore p = \int 2y \, dy = y^2 + A$$

$$\therefore \frac{du_2}{dx} = -2xy + \frac{dq}{dx} = 3x^2 - 2xy \quad \therefore q = \int 3x^2 \, dx = x^3 + B$$

$$\therefore u = x^3 + y^2 - x^2y + A + B = x^3 + y^2 - x^2y + C$$

Question 7: Find general implicit solutions of the following inexact first-order ODEs. Clue: in both cases the appropriate integrating factor is a function of either x or y alone.

$$(a) \frac{dy}{dx} = \frac{x - y^2}{2y}$$

$$(b) (3xy - 6x^2) \frac{dy}{dx} = (6xy - y^2)$$

$$(a) f = x - y^2, g = -2y \quad \therefore \frac{df}{dy} = -2y, \frac{dg}{dx} = 1 \quad \therefore \frac{df}{dy} \neq \frac{dg}{dx}$$

$$\text{Let } I = e^x$$

$$fI = xe^x - y^2e^x, gI = -2ye^x \quad \therefore \frac{dfI}{dy} = -2ye^x, \frac{dgI}{dx} = -2ye^x$$

$$\therefore \frac{dfI}{dy} = \frac{dgI}{dx}$$

$$\text{Let } \frac{du_1}{dx} = fI, \frac{du_2}{dy} = gI$$

$$u_1 = \int fI dx = \int xe^x - y^2e^x dx = xe^x - e^x - y^2e^x + p$$

$$u_2 = \int gI dy = \int -2ye^x dy = -y^2e^x + q$$

$$\therefore \frac{du_1}{dy} = -2ye^x + \frac{dp}{dy} = -2ye^x \quad \therefore p = A$$

$$\therefore \frac{du_2}{dx} = -y^2e^x + \frac{dq}{dx} = xe^x - y^2e^x$$

$$\therefore q = \int xe^x dx = xe^x - e^x + B$$

$$\therefore u = xe^x - e^x - y^2e^x + C$$

Question 7: Find general implicit solutions of the following inexact first-order ODEs. Clue: in both cases the appropriate integrating factor is a function of either x or y alone.

$$(a) \frac{dy}{dx} = \frac{x - y^2}{2y}$$

$$(b) (3xy - 6x^2) \frac{dy}{dx} = (6xy - y^2)$$

$$(b) f = 6xy - y^2 \quad \therefore \frac{df}{dy} = 6x - 2y$$

$$g = 6x^2 - 3xy \quad \therefore \frac{dg}{dx} = 12x - 3y \quad \therefore \frac{df}{dy} \neq \frac{dg}{dx}$$

$$\text{Let } I = y$$

$$\therefore fI = 6xy^2 - y^3 \quad \therefore \frac{d(fI)}{dy} = 12xy - 3y^2$$

$$\therefore gI = 6x^2y - 3xy^2 \quad \therefore \frac{d(gI)}{dx} = 12xy - 3y^2 \quad \therefore \frac{d(fI)}{dy} = \frac{d(gI)}{dx}$$

$$\therefore u_1 = \int fI dx = 3x^2y^2 - xy^3 + p$$

$$u_2 = \int gI dy = 3x^2y^2 - xy^3 + q$$

$$\therefore \frac{du_1}{dy} = 6x^2y - 3xy^2 + \frac{dp}{dx} = 6x^2y - 3xy^2 \quad \therefore p = A$$

$$\therefore \frac{du_2}{dx} = 6xy^2 - y^3 + \frac{dq}{dx} = 6xy^2 - y^3 \quad \therefore q = B$$

$$\therefore u = 3x^2y^2 - xy^3 + C$$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{dN}{dt} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t , α is a positive constant, and m is a positive constant known as the ‘carrying capacity’. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \rightarrow \infty$. Does this make sense?

$$\int \frac{1}{(m-N)N} dN = \int \alpha dt$$

$$\text{For } \int \frac{1}{(m-N)N} dN, \int \frac{1}{(m-N)N} dN = \int \frac{A}{m-N} - \frac{B}{N} dN$$

$$\therefore AN - Bm + BN = 1 \quad \therefore B = -\frac{1}{m} \quad \therefore A = \frac{1}{m}$$

$$\therefore \int \frac{1}{m^2 - mN} dN + \int \frac{1}{mN} dN = \alpha t + A$$

$$\therefore \int \frac{1}{m-N} dN + \int \frac{1}{N} dN = \alpha t + A$$

$$\therefore -\ln|m-N| + \ln|N| + B = \ln\left|\frac{N}{m-N}\right| + B = \alpha t + A$$

$$\therefore \left|\frac{N}{m-N}\right| = e^{\alpha t} + C \quad \therefore \frac{N}{m-N} = e^{\alpha t} + C$$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{dN}{dt} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t , α is a positive constant, and m is a positive constant known as the 'carrying capacity'. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \rightarrow \infty$. Does this make sense?

$$\therefore \left| \frac{N}{m} - 1 \right| = e^{m\alpha t} + C \quad \therefore |N - m| = me^{m\alpha t} + C$$

$$\text{When } \alpha = 1, m = 1000 \quad \therefore |N - 1000| = 1000e^{1000t} + C$$

$$\text{When } N(0) = 10 \quad \therefore C = -10$$

$$\text{When } N(0) = 1200 \quad \therefore C = -800$$

$$\text{As } t \rightarrow \infty, |N - 1000| \rightarrow \infty \quad \therefore N \rightarrow \infty$$

Which not make sense

$$\therefore N = me^{m\alpha t} + mC - Ne^{m\alpha t} - NC \quad \therefore N(1 + e^{m\alpha t} + C) = me^{m\alpha t} + mC$$

$$\therefore N = \frac{me^{m\alpha t} + mC}{1 + e^{m\alpha t} + C}$$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{dN}{dt} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t , α is a positive constant, and m is a positive constant known as the ‘carrying capacity’. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \rightarrow \infty$. Does this make sense?

$$\text{When } \alpha = 1, m = 1000 \quad \therefore N = \frac{1000 e^{1000t} + 1000C}{1 + e^{1000t} + C}$$

$$\text{When } N(0) = 10 \quad \therefore C = -\frac{98}{99}$$

$$\text{When } N(0) = 1200 \quad \therefore C = -700$$

$$\text{As } t \rightarrow \infty, N \rightarrow 1000 \quad \therefore N \rightarrow m$$

Which make sense